

KONGU ENGINEERING COLLEGE · CSE & ECE

System Architecture & Workflow

Smart Construction Site Safety Monitor

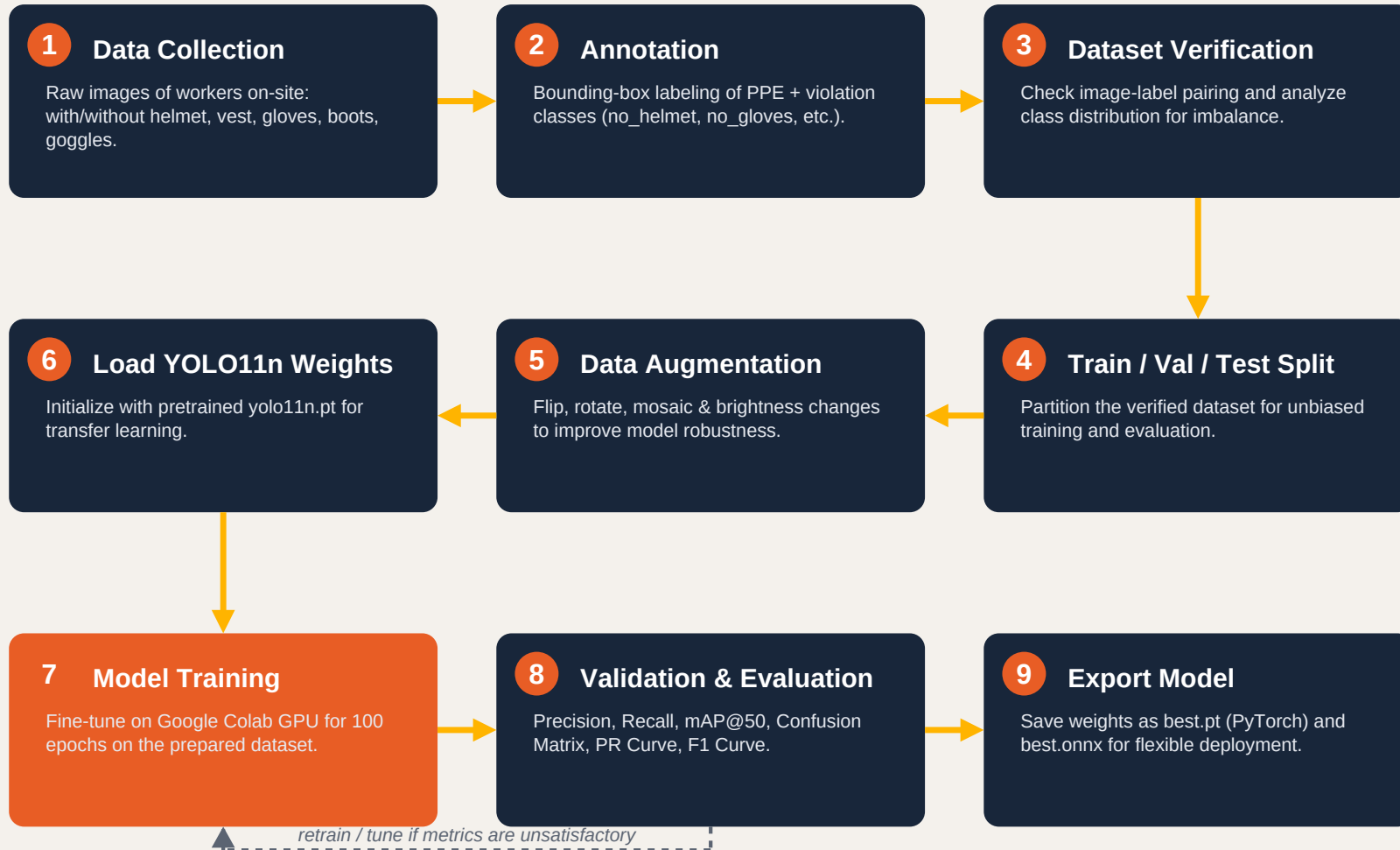
Workflow 1: Dataset & Model Training Pipeline

Workflow 2: End-to-End Project / System Workflow

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Workflow 1 — Dataset & Model Training Pipeline

Google Colab (GPU) · Ultralytics YOLO11 · from raw images to a deployable model

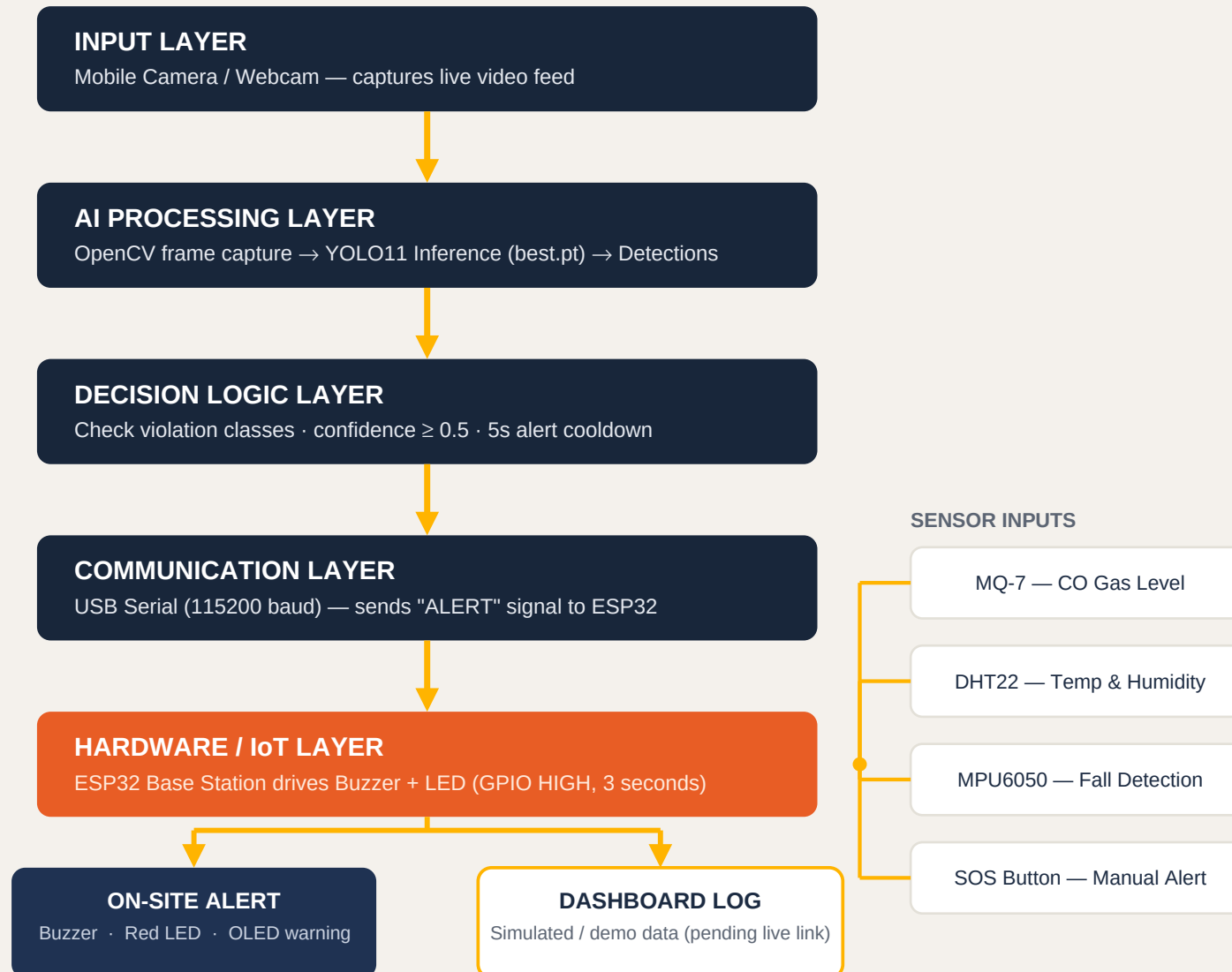


Key Run Details: Platform: Google Colab (GPU) · Base Model: yolo11n.pt · Epochs: 100

Evaluation: Precision · Recall · mAP@50 · Confusion Matrix · PR Curve · F1 Curve · Export: best.pt + best.onnx

Workflow 2 — End-to-End Project Workflow

From live camera feed to on-site physical alert, running the trained model in real time



Note: Workflow 1 runs once, offline, to produce best.pt. Workflow 2 runs continuously, live, loading that trained model to monitor the site in real time.

Smart Helmet — Sensors, Fall Detection & Location Tracking

What the ESP32 helmet unit senses today, and how location tracking fits in compactly

SMART HELMET UNIT — ESP32 Controller

Everything below is mounted on or wired into a single helmet-worn unit

MPU6050

Accelerometer + Gyroscope

Detects falls & abnormal movement — triggers instant alert even if the worker is off-camera.

INCLUDED

MQ-7

Carbon Monoxide Sensor

Monitors CO levels around the worker — critical in enclosed or poorly ventilated areas.

INCLUDED

DHT22

Temperature & Humidity

Tracks heat stress conditions so extreme site temperatures can trigger a warning too.

INCLUDED

SOS Button

Manual Trigger

Lets a worker manually raise an emergency alert independent of automatic sensing.

INCLUDED

GPS Module (NEO-6M)

Real-Time Location

Adds the worker's live position, so an alert tells supervisors WHO and WHERE — not just that something happened.

PROPOSED ADD-ON

WIRELESS TRANSMISSION

Wi-Fi or LoRa from the ESP32 — sends sensor + location data without a wired tether.

BASE STATION

Receives helmet data, drives on-site alerts, and forwards worker status + location to the dashboard.

ON-SITE ALERT

Buzzer, LED, OLED on the helmet / base unit

DASHBOARD

Worker ID, status & live location on a site map

Feasibility: A GPS module like the NEO-6M is roughly coin-sized and low-power — it connects to the same ESP32 over a spare UART/I2C line as the existing sensors, so it adds location tracking without changing the helmet's size or design.